Multimedia Application & Virtual Reality

X3D

🌐 What is X3D?

✨ **X3D = Extensible 3D**, the modern ISO standard for interactive graphics

🌍 Successor of **VRML**, designed for the modern web

🧩 Built on a **scene** graph with **nodes**, **fields**, and **events**

🚀 Used in **VR**, **AR**, **CAD**, simulation, scientific visualization

🎯 Scope of X3D

- 🌐 **Web-based** 3D graphics
- 🥽 **Interactive** VR/AR environments
- 🏗️ CAD and engineering
- 🧪 Education, medicine, and simulations
- 🔌 Supports **XML**, Classic **VRML**, and **JSON** syntaxes

🎱 X3D Architecture

- 🧩 **Nodes** as individual components
- 🔗 **ROUTEs** connecting events
- 🖱️ **Sensors** for user interaction
- 🕒 **Interpolators** for animation
- 🧠 **Scripts** for logic
- 🎨 Appearance, textures, and shaders

🧩 Node Types

- 🗂️ **Grouping** nodes (Transform, Group)
- 🔷 **Geometry** (Box, Sphere, IndexedFaceSet)
- 🎨 **Appearance** (Material, ImageTexture)
- 🖱️ **Sensors** (TouchSensor, PlaneSensor)
- 🕒 **Animation** (TimeSensor, Interpolators)
- 💡 **Lights** (PointLight, DirectionalLight)

Encodings of X3D

- **XML** format
- Classic **VRML** format
- **JSON** format
- All represent the same scene graph

Basic X3D Scene Structure (Concept)

- An **X3D scene** always contains:
  - The `<X3D>` root
  - A `<Scene>` element
  - Shapes, transforms, sensors…

Basic X3D Scene Structure Example

```
<X3D>
    <Scene>
        <Shape>
            <Appearance>
            <Material diffuseColor='0 0 1' />
            </Appearance>
            <Box />
        </Shape>
    </Scene>
</X3D>
```

🔷 Shapes (Concept)

- 🎲 Box
- ⚽ Sphere
- 🟪 Cylinder
- 🔺 Cone
- 🧱 IndexedFaceSet for custom geometry

Shapes Example

```
<Scene>
    <Shape>
    <Appearance>
        <Material diffuseColor='1 0 0'/>
    </Appearance>
    <Sphere />
    </Shape>
</Scene>
```

🎨 Appearance & Material (Concept)

- 🎨 **Material** controls object color
- 💡 Defines ambient, diffuse, and specular light behavior
- 🌟 **Shininess** affects highlights

Appearance & Material Example

```
<Shape>
    <Appearance>
        <Material diffuseColor='0 1 0' shininess='0.8' />
    </Appearance>
    <Box />
</Shape>
```

🔁 Transformations (Concept)

↔ translation

🔄 rotation

📏 scale

🧪 Transformations (Executable Example)

```
<Transform translation='0 1.5 0' rotation='0 1 0 1.57'>
    <Shape>
        <Cylinder />
    </Shape>
</Transform>
```

🖱️ Sensors (Concept)

- 👆 **TouchSensor** detects clicks
- 🖐️ **PlaneSensor** detects dragging
- 🔁 **CylinderSensor** detects rotation
- 🎯 **ProximitySensor** detects viewer location
- 👁️ **VisibilitySensor** detects visibility

TouchSensor Example

```
<Shape DEF='OBJ'>
    <Appearance>
        <Material diffuseColor='0 0 1' />
    </Appearance>
    <Box />
</Shape>
<TouchSensor DEF='CLICKER' />
```

🕒 Animation & TimeSensor (Concept)

- 🕒 **TimeSensor** generates continuous time
- 🎞️ **Interpolators** convert time into movement
- 🔗 **ROUTEs** connect animation nodes

Simple Animation Example

```
<Transform DEF='MOVE_ME'>
    <Shape>
    <Box />
    </Shape>
</Transform>
<TimeSensor DEF='CLOCK' cycleInterval='3' loop='true' />
<PositionInterpolator DEF='PATH'
key='0 0.5 1'
keyValue='0 0 0  0 2 0  0 0 0' />
<ROUTE fromNode='CLOCK' fromField='fraction_changed' toNode='PATH'
toField='set_fraction' />
<ROUTE fromNode='PATH' fromField='value_changed' toNode='MOVE_ME'
toField='set_translation' />
```

🖱️ 🎞️ Sensor & Animation on Cube

- 👆 **TouchSensor** triggers animation
- 🕒 **TimeSensor** controls timing
- 🎞️ **Interpolator** moves the cube
- 🔗 **ROUTEs** connect all components

Sensor + Animation Example

```
<Scene>
    <Transform DEF='CUBE'>
        <Shape>
            <Appearance>
                <Material diffuseColor='0.2 0.7 1' />
            </Appearance>
            <Box />
        </Shape>
    </Transform>
    <TouchSensor DEF='CLICK' />
    <TimeSensor DEF='TIMER' cycleInterval='2' loop='false' />
    <PositionInterpolator DEF='ANIM'
    key='0 1'
    keyValue='0 0 0  0 3 0' />
    <ROUTE fromNode='CLICK' fromField='touchTime' toNode='TIMER' toField='startTime' />
    <ROUTE fromNode='TIMER' fromField='fraction_changed' toNode='ANIM' toField='set_fraction' />
    <ROUTE fromNode='ANIM' fromField='value_changed' toNode='CUBE' toField='set_translation' />
</Scene>
```

🌐 X3D in HTML (Concept)

- 🧩 X3D can be **embedded directly** in web pages
- 🌍 Works with **X3DOM** or **X_ITE**

HTML Integration Example

```
<x3d width='500px' height='400px'>
    <scene>
        <shape>
            <box></box>
        </shape>
    </scene>
</x3d>
```

X3D for VR/AR

- Compatible with **WebXR**
- Supports **device sensors**
- **Viewpoints** enable head-tracking

🧭 Summary

- 🌟 **Modern** 3D standard
- 🧱 **Node-based** architecture
- 🖱️ **Sensors** for interaction
- 🎞️ **Animation** via TimeSensor and Interpolators
- 🧠 **Extensible** with scripts
- 🌐 **Ready** for Web, VR, and AR